

realized it was a general phenomenon. Rubin's work convinced astronomers that dark matter exists. She received many prestigious awards, including the National Medal of Science in 1993.

PETER HIGGS

1929–

Born in Newcastle, England, Peter Higgs studied physics at King's College London. In 1960, he began work at the University of Edinburgh. Higgs is best known for his involvement in developing a theory that explains the origin of the mass of elementary particles (subatomic particles). According to this theory, these particles obtain their masses by interacting with a field, called the Higgs field, that permeates space. He predicted that this field should produce its own type of particle, which has since become known as the Higgs boson. The search for the Higgs boson spawned the world's largest science project, the Large Hadron Collider (LHC) at CERN, and scientists here went on to discover a particle with the predicted characteristics of the Higgs boson in 2012. A year later, Higgs received the Nobel Prize in Physics, sharing it with François Englert for the work they carried out in 1964.

ERIC KANDEL

1929–

Born in Vienna in 1929, Eric Kandel fled with his family to the US as anti-Semitism became rife. He was interested in the cellular basis of behavior and studied medicine at New York University, receiving his degree in 1956. In 2000, along with two others, he was awarded the Nobel Prize in Physiology or Medicine for work on the role synapses (nerve junctions) play in memory and learning. Through investigations on the

sea slug (which has a simple nervous system with large cells that are easy to study), he helped to clarify the different processes behind short- and long-term memory and showed that they are equally applicable in humans.

MARIO J. MOLINA

1943–

Born in Mexico City, Mario Molina studied to become a chemical engineer. In 1974, he and colleague Sherwood Rowland showed that CFCs (chlorofluorocarbon compounds) released from aerosols, foams, and refrigerants destroy the ozone layer, which helps protect humans from the Sun's ultraviolet radiation. This work led directly to banning or limiting the use of CFCs and earned Molina and Rowland a share of the 1995 Nobel Prize in Chemistry.

LAP-CHEE TSUI

1950–

Canadian medical geneticist Lap-Chee Tsui was born in Shanghai and moved to Canada in 1977. He jointly found the fault in the gene responsible for cystic fibrosis. The life-threatening disease causes mucus produced by the body to be very thick—the mucus particularly affects the lungs. The identification of the faulty gene paved the way for new treatment options based on correcting the gene or replacing it.

MAE CAROL JEMISON

1956–

The first African American woman to travel in space, Mae Carol Jemison went into orbit in the space shuttle *Endeavour* in 1992. Jemison had graduated from medical school in 1981 before serving

as a medical officer with the Peace Corps in West Africa. In 1985, she applied to the US space agency, and in 1987, she became one of the 15 candidates selected from more than 2,000 applicants to train as an astronaut. Jemison was the science mission specialist on the space shuttle and a coinvestigator of two bone cell research experiments performed there.

DONNA STRICKLAND

1959–

Canadian optical physicist Donna Strickland paved the way for the shortest and most intense laser pulses ever created. In 1985, along with Gérard Mourou from France, she developed a technique called chirped pulse amplification (CPA), and for this she and Mourou were jointly awarded the Nobel Prize in Physics in 2018. Strickland became associate professor in the Department of Physics and Astronomy at the University of Waterloo in 1997, where she now leads the ultrafast laser research team. CPA has several uses, including in laser eye surgery and in some cancer therapies.

CAROL GREIDER

1961–

Most of American molecular biologist Carol Greider's research has been on telomeres and the enzyme telomerase. Telomeres are the segments of DNA that occur at the ends of chromosomes and play an important role in cell life span. In 2009, along with molecular biologist Elizabeth Blackburn and geneticist Jack W. Szostak, Greider was awarded the Nobel Prize in Physiology or Medicine. She found that inhibiting the telomerase in cancer cells slows tumor growth, making it a potential anticancer drug development.